

AVOCADO PEST MANAGEMENT



Venue: Kitale National Polytechnic

Presenter: Kennedy Muteti,

KEPHIS



OUTLINE

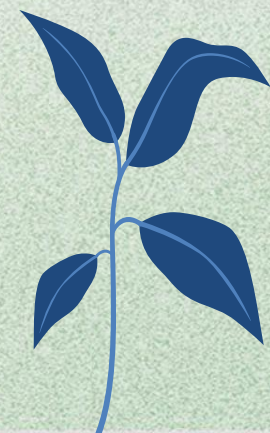
- ✓ **Persea Mites**
- ✓ **Fruit flies**
- ✓ **FCM**
- ✓ **Scale insects**
- ✓ **Thrips**





Persea Mites (*Oligonychus perseae*)

Taxonomic classification

- **Kingdom:** Animalia → all animals
 - **Phylum:** Arthropoda → joint - legged animals with exoskeleton
 - **Class:** Arachnida → 8 - legged arthropods (spiders, ticks, mites)
 - **Order:** Trombidiformes → specific group of mites, usually very small, often microscopic.
 - **Family:** Tetranychidae → spider mites
 - **Genus:** *Oligonychus* → plant-feeding mites
 - **Species:** *Oligonychus perseae* → Persea mite
- 

Persea Mites (*Oligonychus perseae*)

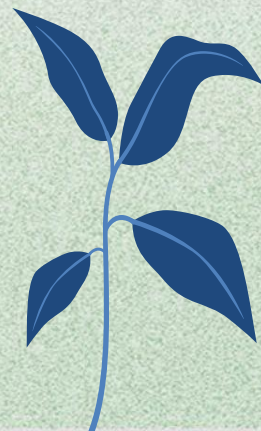
Literature Review

Persea mites are small, sap-sucking arachnids that infest the underside of avocado leaves, where they puncture plant cells and extract contents. They form clusters protected by silk webbing, especially along major veins and the leaf midrib.



Their feeding results in:

- ❑ **Chlorotic and necrotic lesions** that reduce photosynthetic capacity
- ❑ **Premature leaf senescence** and drop
- ❑ **Exposed fruits**, increasing risk of sunburn and reduced market value



Literature Review, Cont....

- ▶ The pest was **first reported in Mexico and California**, where it **caused severe economic losses**. It has since spread to other avocado-growing regions globally. Its **introduction in Kenya which was first reported in Nakuru County**, is suspected to be through **infected scions or seedlings**, and its movement is facilitated by **wind, human activities, and poor nursery practices**.
- ▶ Varietal susceptibility varies. The **Hass variety**, widely grown for export, has been documented in international studies **as highly susceptible**. Since Kenya's avocado export market is dominated by Hass, the pest represents a significant phytosanitary risk.



Literature Review, Cont....

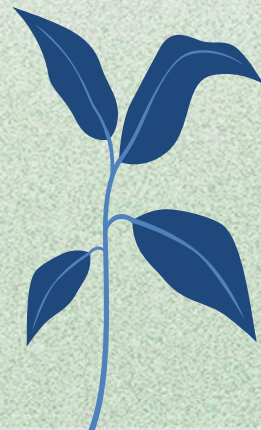
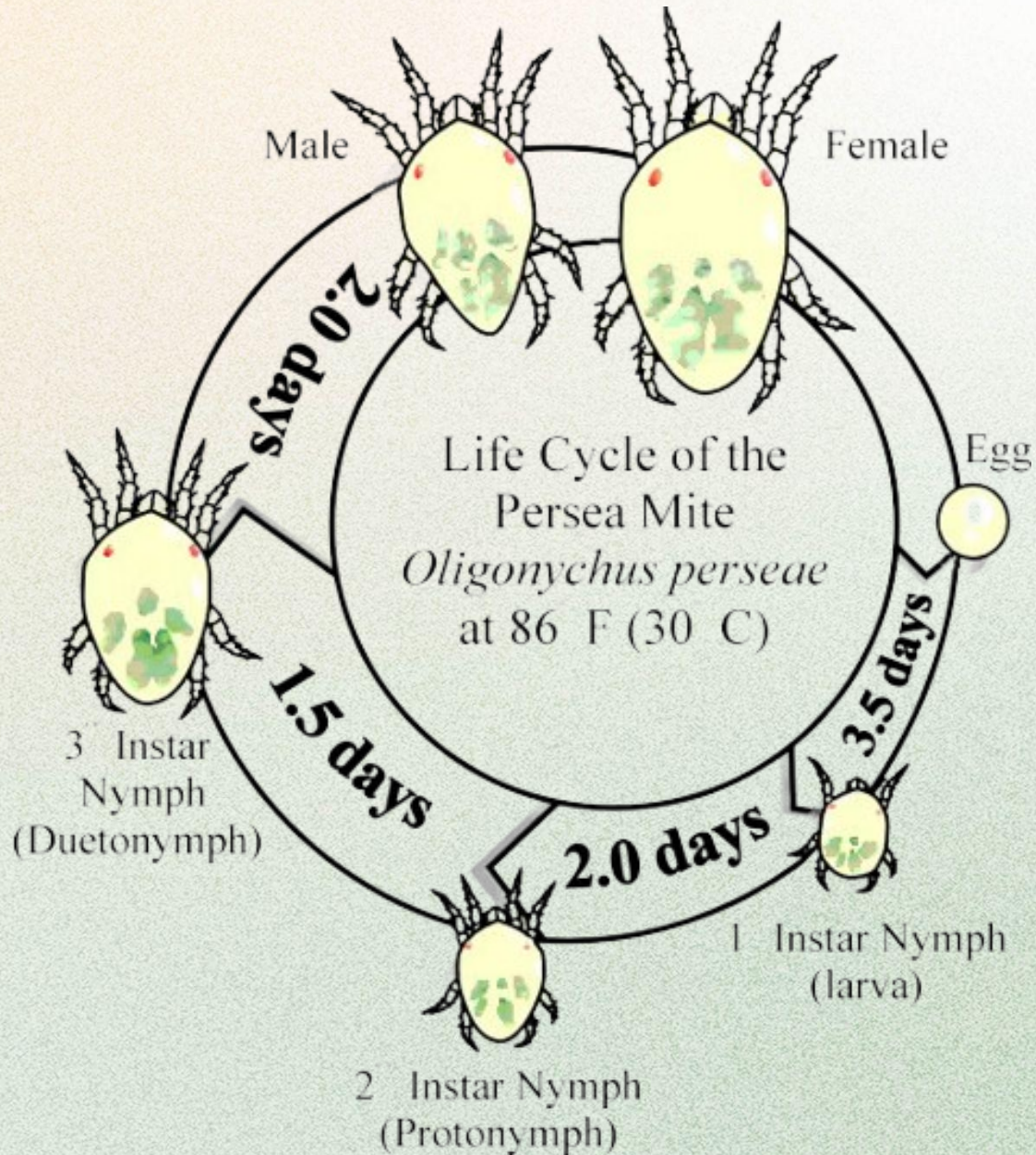
- ▶ Effective management of Persea mites requires an Integrated Pest Management (IPM) approach, combining regular scouting, cultural practices (e.g., pruning, sanitation), conservation of natural enemies (e.g., predatory mites), and the judicious use of acaricides.
- ▶ However, **low awareness among farmers and limited pest surveillance** has contributed to underreporting and misdiagnosis in the country.





Persea Mites

Life Cycle





5 Developmental stages:

- Egg
- Larva
- Protonymph
- Deutonymph
- Adult

Adult

- 8 legs
- Most destructive
- Females - 0.5 mm long, oval, yellowish-green with dark spots
- Males – smaller & pear-shaped.

Developmental Rate

- Egg to adult takes approximately 15 days at 30°C and up to 40 days at 20°C
- With populations increasing during warmer months





Persea Mites: Behavior & Traits

Reproduction

Persea mites reproduce by laying 20 to 45 eggs per female during her life.



Feeding Habits

They feed in colonies on the underside of leaves, piercing tissue to suck sap, causing circular yellow to dark brown necrotic spots, leaf wilt, and defoliation, which can reduce fruit size and yield.

Location & Webbing

Mites create protective webbing on the underside of leaves along the midrib and veins, appearing as silvery spots.



Damage & Manifestation

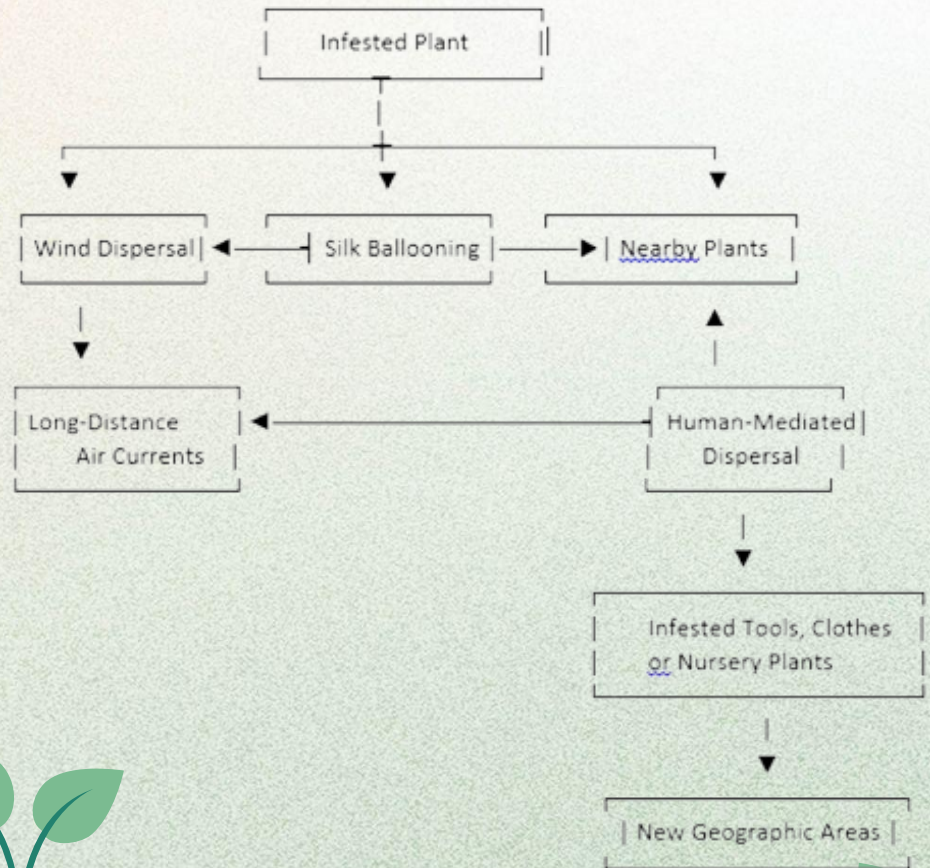
Severe infestations lead to leaf yellowing or browning and eventual leaf drop, which can be mistaken for fungal infections.



Pathways of Dispersal

Primary Dispersal

Wind Assisted by silk strands or crawling



Pathways of Dispersal

Secondary Dispersal

Human activities Harvesting, pruning, weeding, spraying

Infested plant material,

Farm tools and equipment
(Hitch hiking)

The mites are not normally transported through fruits and this minimizes the risk of fruit transmission.



The Key source of transmission is through planting materials and cuttings. Examples: from Mexico to California was through illegal transfer of avocado cuttings

Pathways of Dispersal

- Infested mother-plants (scions)
- Nursery seedlings
- Imported planting material
- Buyers movement between farms
- International, regional and domestic trade



Population Dynamics

Populations
fluctuate cyclically.



Influenced by
temperature,
humidity, food
availability, and
natural enemies



The population
drops in cold,
rainy seasons and
increase in warm,
humid, windy
conditions



Damage



They cause damage by feeding on the underside of young leaves



The mites feed close to the veins forming a characteristic damage pattern on the leaves



Damaged leaves turn yellow, brown or get holes between the veins



Signs and Symptoms

Circular Spots

Circular, yellow to dark brown necrotic spots

Scattered along veins and midribs

On the underside of avocado leaves, visible through the upper surface



Effects of Infestation



Defoliation



Sunburn

Defoliation and Sunburn

- Heavy Persea mite infestation causes defoliation
- Exposure to direct sunlight leads to sunburned bark and fruits.



Effects of Infestation



Stress & Yield Reduction

Affected trees appear stressed, resulting in

- ✓ Premature fruit ripening,
- ✓ Flower drop
- ✓ Fruit drop
- ✓ Reduced yields.

Host Range



Avocado



Peach



Castor Beans



Plum



Roses



Eucalyptus



Passion Fruit

Scouting



- Persea mite monitoring requires early detection
- Scout every 7-10 days, observe both upper and underside of leaves for signs and symptoms.
- Confirm using
 - ✓ a **hand lens**,
 - ✓ **exposing the leaf underside to direct sunlight**
 - ✓ **shaking sampled leaves onto white paper**

Cultural Control of Persea Mites

Regular Scouting & Monitoring

Management should begin when 7.5 -10% of the leaf surface is damaged or 70 -100 mites per leaf are estimated.



Cultural Control of Persea Mites

Remove alternate hosts

Remove alternate
host plants to
integrate into
management
programs.



Cultural Control of Persea Mites

Use clean planting materials Procure clean, pest-free seedlings from certified nurseries and conduct routine checks to prevent the introduction and spread of mites.



Cultural Control of Persea Mites

Proper spacing & Pruning Maintain recommended spacing (8m x 8m) and pruning to reduce dense canopies, improve aeration, and increase light penetration, hindering mite reproduction and enhancing spray.



Cultural Control of Persea Mites



Sanitation of tools

Clean farm tools and equipment after use in infested areas and limit movement between orchards to prevent mite dispersal on overalls, fruit tins, and harvesting gear.





Biological Control

Predatory Mites (*Neoseiulus californicus*)

Commercially available predatory mites show potential for Persea mite control.

- Eats 5 mites or eggs per day

Release Strategy

- Experiments indicate that manually releasing minimum of 2,000 predators per tree to effectively control Persea mites.
- Conserving natural enemies like ladybirds and beetles, and using entomopathogenic fungi, are also part of biological control efforts.

Organic Plant Extracts

- Extracts from neem can be used to control the mite.
- It is important to spray under the leaf to enhance target area.





Recommended Products (Under Emergency Registration)

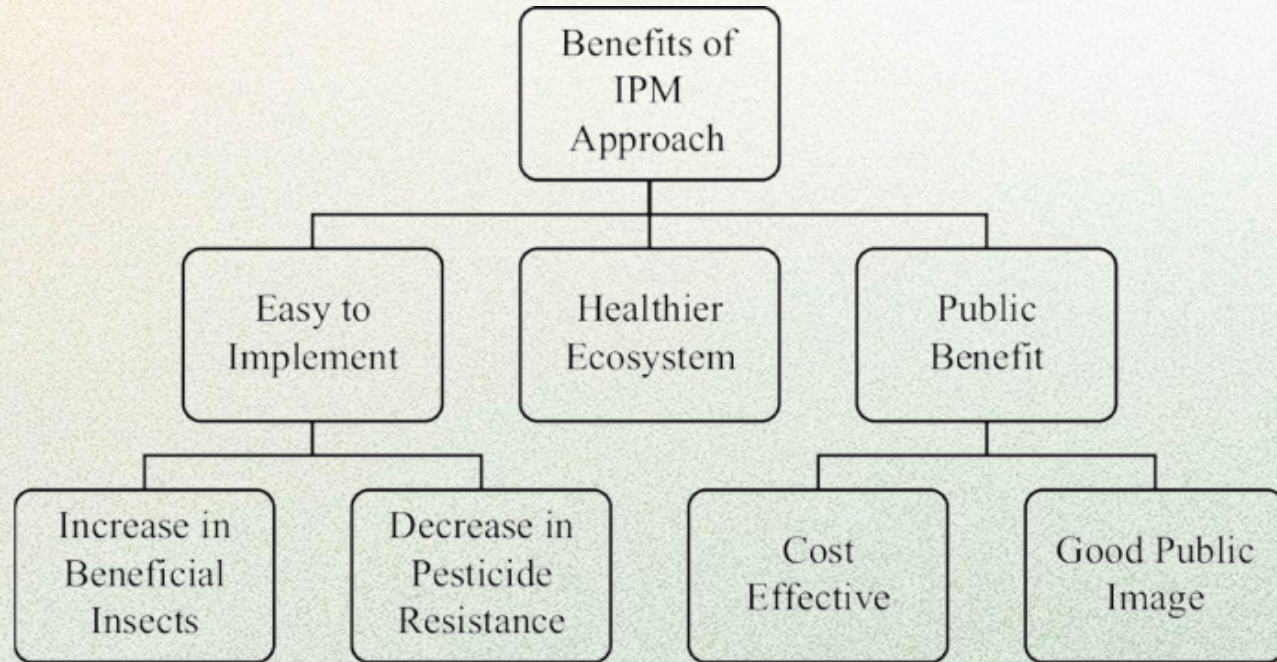
- **Mazao achieve** PCPB(ER)0057 **from Real IPM** is a formulation of *Metarzhium anisopliae* which is a biopesticide.
- **Spidor 240SC** PCPB(CR)1608 from Kenagro Industries, an insecticide that can be used to control Persea Mite.
- **Real carlifonicus** PCPB(ER)0058 **from Real IPM** - Predator mite used in other countries for the control of Persea mite

NB: Refer to PCPB website for comprehensive list of registered / approved pest control products.



Benefits of IPM

- ❖ Consumer Confidence
- ❖ Ensures reliable, stable, and quality avocado yields
- ❖ Decreases the severity of pest infestations.
- ❖ IPM delays pest resistance and resurgence
- ❖ Improves crop profitability
- ❖ Reduces the potential for interceptions/rejection at the market.



KEPHIS 2025 survey findings

Variety Susceptibility

Hass: Most severely affected in all regions, with extensive chlorotic spotting and canopy thinning.

Fuerte and local varieties: Showed limited to mild symptoms. Further studies are needed to establish tolerance levels.

Farmer Awareness

Less than 15% of the farmers had prior knowledge of mites. Many misdiagnosed the damage as nutrient deficiency, drought stress, or sunburn.; Some farmers had used acaricides without proper guidance, while a few relied on traditional practices like neem spray.

Conclusion

The 2025 survey confirms that perseia mite has established itself in Kenya's avocado-producing counties. Without urgent action, its continued spread could compromise both yields and market access, especially for Hass exports.

Key Conclusions:

The mite is more prevalent in warm, dry zones; Hass is highly susceptible and represents the main risk for economic losses. Farmer knowledge and surveillance infrastructure are inadequate; Nurseries may serve as primary spread points.

Major Fruit flies of concern to export markets

C. capitata



C. cosyra



C. rosa

...Q list for export markets



FCM



B. dorsalis



B. cucurbitae

Fruit fly Life-cycle .../1



Monitoring

- Regularly scout when fruit starts setting
- Look for adult fruit flies using fruit fly traps.
- Look out for premature fall of fruits that start to rot.
- Look on fruit surfaces for small holes, punctured fruit, black lesions and soft spots.
- Look inside fruit for rotting, maggots that jump on touch and dis-coloration.
- Take action to control as soon as fruit flies and damage symptoms are observed on one plant.



Orchard sanitation

- Avoid build up of fruit fly populations by removal of fallen fruits.
- To be done regularly (twice a week for the entire season).
- Kill the maggots by burning, burying or trying collected fruits in black plastic bags and exposing them to the heat to kill the maggots,
- When burying fruits, ensure that the fruits are buried at least 50 cm (about two feet) deep to prevent emerging adult flies from reaching the soil surface.



Black plastic bags



Trapping

- Used with either specific attractants such as **methyl eugenol** or **food baits** such as *protein hydrolysates or yeast*
- ME attracts males of *Bactrocera* species and of a few *Dacus* species; Food baits attract both males and female fruit flies, they are not species specific, and also attract other insects, including natural enemies.



False Codling Moth



Caterpillar (left) & adult (right) of the False Codling Moth

36



False Codling Moth

Identification:

- Adult is a **small brownish, night-flying moth**
- Females lay eggs singly, **mostly on fruit**
- Larvae are **white to pinkish color** with **brown to black head**

Symptoms:

- After emerging from the egg, the young larvae **tunnels into the fruit**
- The larvae feed from inside the fruit with **frass** or **excrement** being produced at the entrance of larval tunnels



Larva stages



L2 and L3
larva (white)



L4 larvae
(pink)



L5 larvae (pink)

FCM Larvae



False Codling Moth Management

- Proper **orchard sanitation**
- **Infested fruits** (both on the tree and fallen) **should be removed regularly** and disposed of appropriately
- Manage the false codling moth in other host crops like coffee, guava,
- Scouting
- Trapping- pheromone

FCM Traps



Use of pheromone traps

- Monitoring: Place 1 trap per hectare to detect FCM presence early. Check weekly for moths inside.
- Mass Trapping: Use 5–10 traps per hectare to capture male moths and reduce reproduction.
- Placement: Hang traps at fruit height inside the tree canopy, away from direct sunlight.
- Maintenance: Replace lures every 4–6 weeks and change sticky liners when full.







Management of scale insects

- **Monitoring & Identification**

- Inspect leaves (especially undersides), stems, and fruit for small, immobile bumps.
- Look for signs of infestation: Sooty mold (from honeydew)
- Yellowing leaves
- Ant activity (ants protect scales)



Management of scale insects

- **Reduce plant stress**
 - Scale outbreaks are worse on stressed trees.
 - Ensure **proper irrigation** (avoid water stress).
 - Maintain **balanced fertilization** (excess nitrogen can boost scale populations).
- **Prune to open the canopy**
 - Improves air circulation and spray penetration.
 - Removes heavily infested branches.



Management of scale insects

- **Control ants**
 - Ants protect scale insects from predators.
 - Use **sticky barriers** or **bait stations**.
- Enhance beneficial insects :
 - **Lady beetles** (e.g., *Chilocorus* spp.)
 - **Lacewings**
 - **Predatory mites**
 - **Parasitic wasps** (e.g., *Coccophagus*, *Metaphycus*)
 - Avoid broad-spectrum insecticides that kill natural enemies and can cause scale flare-ups.



Management of scale insects

- **Chemical treatments** are used only when infestations exceed thresholds.
- **Horticultural oils**
 - Smother scales at crawler stage
 - Apply at the correct timing (crawler peak)
- **Insect growth regulators (IGRs)**
 - Such as pyriproxyfen or buprofezin (availability depends on region)



Management of scale insects

- Chemical treatments are used only when infestations exceed thresholds.
- **Systemic insecticides** (only when necessary)
 - e.g., imidacloprid or spirotetramat
 - Effective for severe infestations, but must be used carefully to protect beneficials.
- → **Timing is critical:** Treat during the *crawler stage*, when young scales are mobile and vulnerable.



Management of scale insects

- Post-harvest Considerations (for commercial orchards)
- Avoid moving infested bins and tools between blocks.
- Clean harvest equipment to prevent spreading scales.





Avocado scale insects





Thrips



Photo: Jack Clark, University of California - Davis, Bugwood.org (CC BY 3.0 US)

Avocado thrips



Photo: Mark S. Hoddle, University of California - Riverside, Bugwood.org (CC BY 3.0 US)

Thrip Damage

55



Thrips

Symptoms:

- The infested plant tissues will turn **brown** and **dry up**, if the damage is too severe
- On fruit, feeding begins **near the calyx**, gradually producing **a scar** that can cover the whole fruit
- Attacked fruits develop **a leathery brown skin**



Thrips Management

- Regular monitoring by scouting and use of sticky traps
- Conserve natural enemies: Thrips are attacked by **predatory Thrips, Lacewings and Predatory Bugs**
- Use of appropriate pesticides
- Tree management- Feeding, pruning



Coloured sticky traps







AVOCADO DISEASES & MANAGEMENT



Major Diseases

- The following are the major diseases of Avocado in Kenya:
 - a) Root Rot
 - b) Sunblotch
 - c) Anthracnose
 - d) Cercospora Fruit Spot
 - e) Scab



Avocado Root Rot



Phytophthora Root Rot

General Information:

- This is a **fungal disease**
- It is **the most serious disease** in nearly all Avocado producing areas of the world and in Kenya
- In areas subject to **flooding** and in **poorly drained soils**, trees of any age or size are likely to be infected by the fungus
- Trees affected **lose vitality, become sparsely foliated** and do not produce a good crop



Symptoms

➤ Declining tree canopy

- ❖ Leaves are small, pale green, often wilted with brown tips, and drop readily.
- ❖ Shoots die back from the tips
- ❖ Branches usually become **sunburned** and **die** back in advanced stages
- ❖ Eventually the tree is reduced to a bare framework of dying branches



Symptoms

- Small feeder root decline
- ❖ Healthy roots are creamy-white in colour and in abundance
- ❖ Roots black, brittle and decayed
- ❖ Roots absent
- Trees affected **lose vitality, become sparsely foliated** and do not produce a good crop
- Tree death over a few months to several years, depending on soil characteristics, cultural practices and environmental conditions



Factor affecting root rot infection

➤ Site problems

- ❖ Poor drainage
- ❖ Clay soils
- ❖ Hard-pan
- ❖ High water table
- ❖ Planting on steep slopes without terracing



Factor affecting root rot infection

- **Diseased seedlings**
- **Infested irrigation water; soil**
- **Cultural**
 - ❖ Damage to feeder roots through cultivation or other physical damage
 - ❖ Poor irrigation – over watering; watering at the tree trunk instead of drip-line
 - ❖ Growing crops under the avocado trees



Management - Integrated Pest Management

➤ Prevention

- ❖ Selection of appropriate site
- ❖ Modify site to facilitate drainage- ridging, terracing, construct drains to direct water away from the avocado field
- ❖ Hot water or fungicide treatment of rootstock seeds
- ❖ Use seedlings from a certified nursery
- ❖ Grafting on Phytophthora-tolerant and/or resistant rootstocks



Management

➤ Cultural Practices:

- ❖ Provide favorable soil conditions
- ❖ Use certified disease-free nursery stock
- ❖ Plant resistant rootstocks
- ❖ Prevent soil or water movement from infested areas
- ❖ Irrigate carefully, not too much water
- ❖ Apply agricultural gypsum and mulch



Management

➤ Provide appropriate nutrition

- ❖ Moderate amounts of nitrogen promote good growth that helps avocado better tolerate root rot
- ❖ Avoid excess amounts of fertilizer, especially large amounts of animal manures or other products high in ammonia or salts
- ❖ Avocado roots are sensitive to ammonia and salts



Management

- Use high quality irrigation water
- Irrigation water with high overall salinity or an excess of boron, chloride, or sodium promotes infection of roots by *Phytophthora*
- Irrigation water treatment -*Phytophthora* can contaminate irrigation water- e.g. surface water that is runoff from infested soil
- Apply agricultural gypsum and mulch
(Gypsum supplies calcium, which suppresses the formation of *Phytophthora* spores)
- Create soil conditions that suppress development of *Phytophthora* root rot



Management

➤ Chemical Control

- ❖ Certain phosphonate fungicides (phosphorous acid and phosphonate compounds) can improve trees' ability to tolerate, resist, or recover from infection
- ❖ Phosphonates cannot eradicate Phytophthora from the grove and Phytophthora root rot requires ongoing management throughout the life of the trees.
- ❖ Metalaxyl (Ridomil R) can be applied as granular, a drench or with irrigation water
- ❖ Biological control- there are biocontrol agents that can control phytophthora to some extent



Avocado sunblotch viroid disease

- Caused by a viroid; Avocado sunblotch viroid (ASBVd)
- Narrow host range *Persea americana* species the only natural host
- Distribution: Mexico, Guatemala, Peru, Venezuela, USA, Spain, Greece, Israel, Australia, South Africa, Ghana and **Kenya**
- Three strains exist
- It's a quarantine pest in some countries
- Symptoms may appear 36 months after grafting, some trees may be symptomless
- Symptoms depend on host cultivar, age of plants, environmental conditions and possibly the predominance of specific viroid sequence variants
- **Impact:**
 - ✓ Yield loss -15- 80%;
 - ✓ Reduced fruit quality ;
 - ✓ Limited market access



Disease identification



**Yellow streaks
on leaves**



**Discoloured and
deformed foliage**

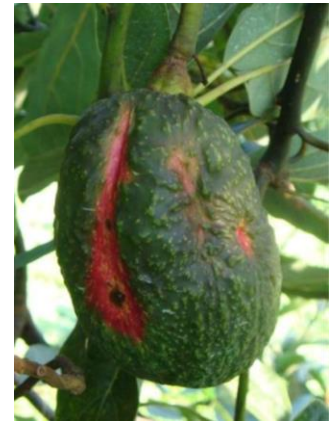


Bleached collar

Disease identification



Declined
canopy



Bleached/ sunken
marks on fruits



Sunburn





Chimera





Bud wood from an ASBVd infected symptomless carrier tree compared to bud-wood from a healthy avocado, the infected bud-wood is larger with bigger buds compared to those from healthy trees.



Disease identification on tree bark



Cracked barks



Transmission

- **Grafting** of unhealthy scion onto a healthy rootstock and vice versa
- **Seed** obtained from an infected tree and used for propagation
- **Pollen** via pollination of a healthy avocado tree by infected pollen
- **Tools** used for pruning and harvesting



Management

- Use of viroid free certified propagation material- scions and seeds
- Testing and indexing of mother block trees
- Removal and destruction of infected trees
- Cleaning and disinfection of tools used for pruning, harvesting and injection equipment between each tree
- Monitoring for timely management
- Screening of imported planting material



Anthracnose

3-

2

0

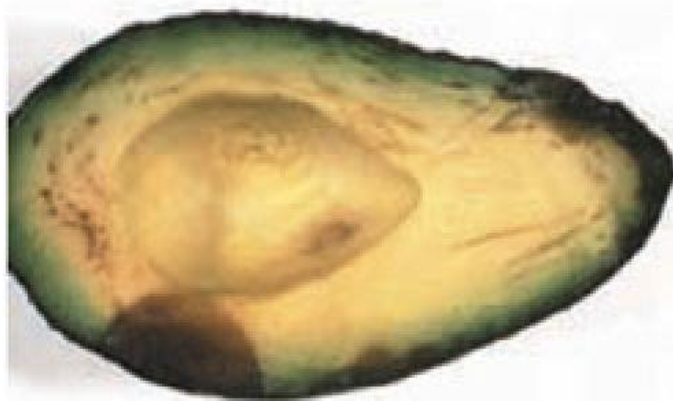


Photo: © A.A. Seif, Icipe
<http://www.infonet-biovision.org/PlantHealth/Crops/Avocados>

Anthracnose on Avocado fruit



Anthracnose



Internally, the rot penetrates deep into the flesh in a hemispherical pattern



Anthracnose

General Information:

- This is a **fungal disease**
- It is important **only in fruits**, where it is the most serious disease affecting the Avocado

Symptoms:

- **Dry spots, dark brown in color**, form on the skin, leading to **abnormal development**
- In severe attacks, **the young fruits drop**
- It is mostly a **post-harvest disease** as the fruits are more susceptible when mature



Anthracnose management

- **Orchard sanitation**
- **Copper-based fungicides** are normally used to control the disease besides other chemicals
- **Pruning**
- **Proper** fertilization- Reduced N increases the fruit shelf life
- Adequate phosphorous and potassium
- Proper irrigation
- Proper drainage



Cercospora Fruit Spot

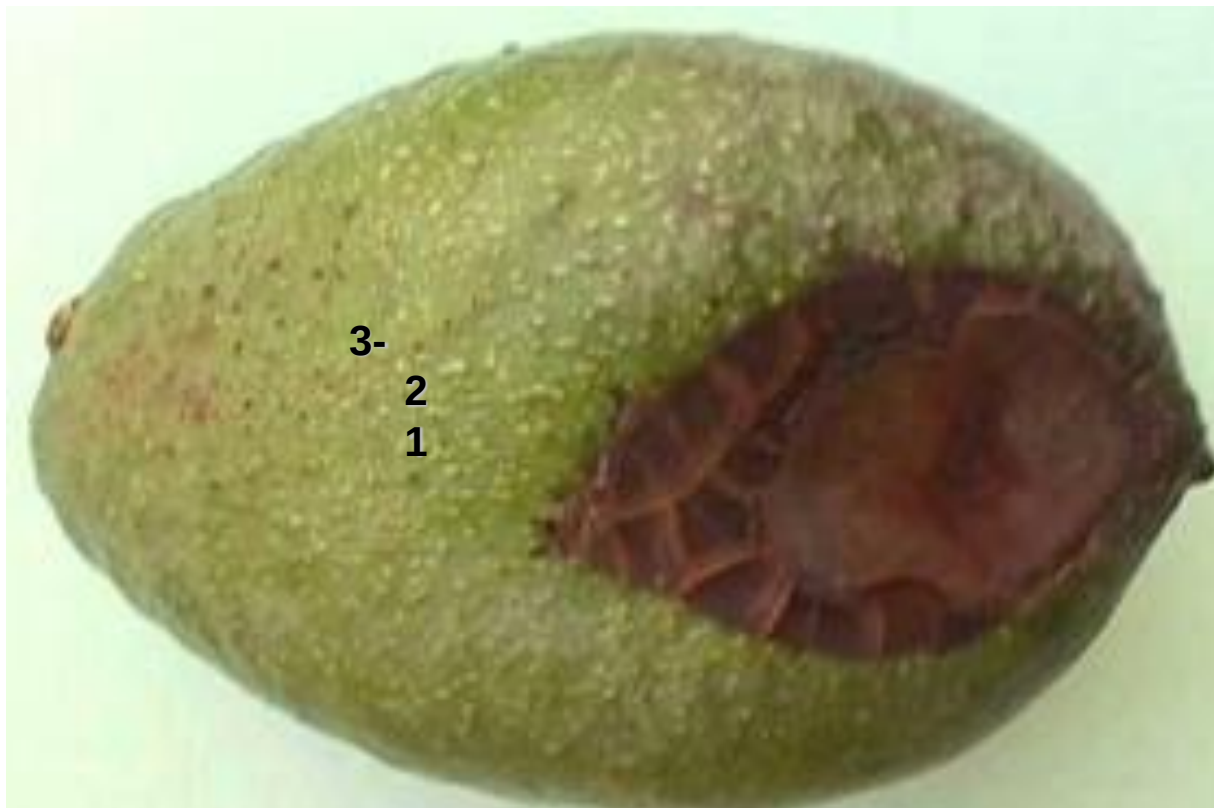


Photo: © A.A. Seif, icipe

<http://www.infonet-biovision.org/PlantHealth/Crops/Avocados>

**Avocado fruit showing a lesion caused
by Cercospora**



Cercospora Fruit Spot

General Information:

- This is a **fungus disease**
- It is **another important fruit disease** impairing the quality of the Avocado
- The severity of infection varies from season to season and can cause **losses of up to 60 %**

Symptoms:

- **Small, light-yellow spots** later changing to **reddish-brown** appear on fruits and leaves which eventually become **hard** and **crack**, thus creating entry points for **other decay organisms**
- **Mature fruits are resistant**



Cercospora Fruit Spot Cont'

Symptoms Cont':

- The disease development is favoured by **humid conditions** and **high temperatures**
- The fungus is essentially spread by **water splash** and its spores are also **wind-borne**

Control:

- **Orchard sanitation**
- **Copper-based fungicides** are normally used (Before flowering and at fruit formation)





Scab

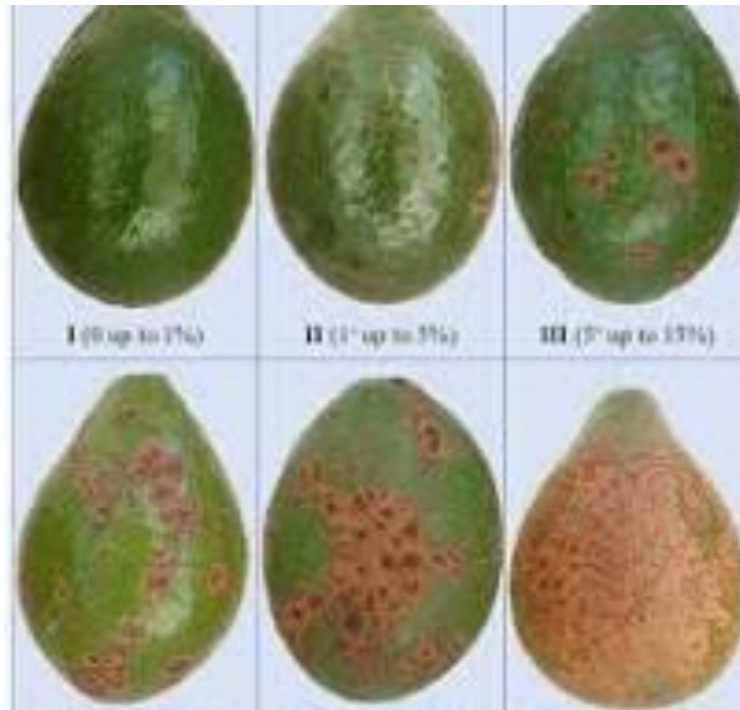
General Information:

- Scab is a **fungal disease**
- It is **common in humid areas**
- The fungus readily infects **young, succulent tissues of the leaves, twigs and fruits**

Symptoms:

- Lesions appear as **small dark spots** slightly raised, oval to elongated
- These coalesce, giving a corky appearance to the surface of the fruits-impairing the appearance **but not the internal quality of the fruit**





Scab Management

- **Orchard sanitation:** remove dead branches and twigs since they harbour the fungus
- Also remove **fallen rotten fruits** from the field
- Apply **copper-based fungicides** at **pre-flowering, fruit formation**
- Healthy trees



Farmer knowledge

- Orchard establishment and management
- Tree management- nutrition, pruning
- Pest monitoring, identification, and management
- Consult Agriculture Extension Officers and other government officers





AVOCADO ORCHARD BIOSECURITY



What is Orchard Biosecurity

- Biosecurity is a set of practices used to prevent, minimize and manage the introduction and spread of pests in the orchard
- Orchard biosecurity is the responsibility of every person working or visiting on your farm
- Every farm requires a biosecurity plan



Critical Areas

1. Pest
2. Product Management
3. People and biosecurity
4. Equipment and Vehicles



1. Pest

- Orchard staff are familiar with common pests and diseases
- Staff know how to detect and identify pests
- Pest scouting regularly conducted, with activities and results recorded even when nothing is found
- Commercial trees and neighbouring vegetation regularly inspected for pests



2. Product Management

- Propagation material thoroughly checked upon receipt
- Records of planting material and its source are maintained
- Staff are aware of symptoms of avocado fruit pests and diseases spread with propagation material
- No soil, plant material or insects left on equipment or tools
- Fallen or waste fruit, packing shed waste and plant trimmings disposed of away from production areas and irrigation sources



3. People

- Staff trained in pest and diseases of avocado
- All visitor and staff clothing, footwear and tools are free of loose soil or plant matter before entering and leaving the orchard
- Footbaths and scrubbing brushes are provided



4. Equipment and Vehicles

- Designated parking area for visitors
- Disinfectant trough for vehicles and footbaths for footwear disinfection
- Machinery entering the property inspected for pests, soil and plant material prior to entering production areas
- Vehicle movement kept to a minimum in production areas
- Pruning, budding and grafting knives are disinfected between seedlings or trees



Thank you for your kind attention!

For more information contact:
Managing Director
KENYA PLANT HEALTH
INSPECTORATE SERVICE
(KEPHIS)

P. O. Box 49592-00100 Nairobi,
Kenya.

Tel: +254-722 891000

E-mail: director@kephis.org

Website: www.kephis.org



Kenya Plant Health Inspectorate
Service-KEPHIS



@KEPHISKE



WWW.KEPHIS.GO.KE

